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Study of the zinc action on the 2-chloroethyl 2-bromo-2-perfluoroalkylethanoates

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ABSTRACT

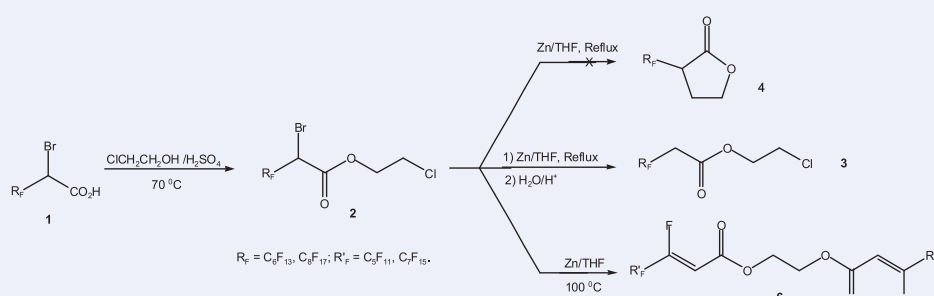
2-Bromo-2-perfluoroalkyl acids are converted into the corresponding esters by a reaction of alcoholysis in the 2-chloroethanol. The action of zinc on perfluoroalkylated bromo-esters resulted in the bromine reduction products formation via zinc organic intermediates. When heated at 100°C, for 48 h, in the presence of an excess of zinc, the bromo-esters produce the symmetrical bis(vinyl perfluoroalkyl ester) ethoxide compounds, resulting from the BrZnF elimination reaction, followed by a dimerization reaction.

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KEYWORDS

Perfluoroalkyl; bromozincic; 2-bromo acid; perfluoroacryl; dimerization



1. Introduction

In recent years, a particular interest is accorded to the synthetic methods of highly fluorinated organic compounds. The perfluoroalkyl chains constitute an important source of reagents for these syntheses. These chains have particular chemical [1,2], physical [3–6] and biological [7–10] properties. Their introduction into organic molecules gives them particular properties compared to their non-fluorinated homologues [11].

The organoperfluoroalkyl compounds have a great number of applications in different fields [12–16]. Such compounds are used as surfactants [17–20], agrochemicals [21,22], pharmaceutical [23–27], biological [28] and medical agents [29,30]. While they are detected in human bodies [31,32], municipal water and solid wastes in different industrial areas around the world [33–35] and considered as high polluter for environment and toxic compounds [36,37], perfluoroalkyl acids are interesting compounds used as tensioactifs [38–40], fire-fighting

foam [41–43], fluoropolymers [44] and other important applications.

In organic synthesis, some perfluoroalkylated compounds can be easily transformed into their corresponding organometallic derivatives [45,46]. However, their chemical reactivity is relatively complex [47]. Many studies describe different positive [48,49] and negative [50] results on their reactivity, adopting different reaction conditions [48–50].

The perfluoroalkyl carboxylic acids are described as starting material for the synthesis of esters [51–53] and heterocyclic compounds [54,55]. In particular, the 2-bromo-2-perfluoroalkyl acids and esters are described in Reformatsky reaction, conducting to the β -hydroxyesters [56–60] and water elimination reaction products [61].

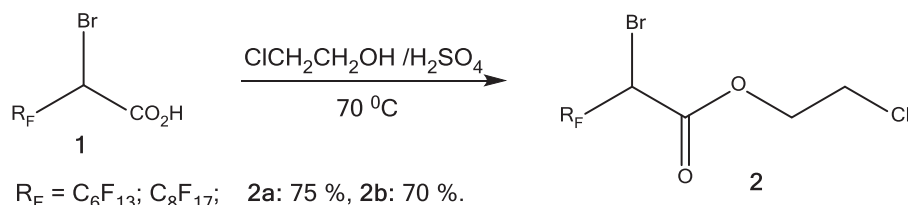
We have previously described the action of zinc and vinyl zinc bromide on the 2-bromo-2-perfluoroalkylethanoic acid and the methyl 2-bromo-2-perfluoroalkylethanoate compounds [52]. As a continuation, we describe herein an attempt of the synthesis of some perfluoroalkylated lactones [54],

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Scheme 1. Alcoholysis reaction of 2-bromo-2-perfluoroalkyl acids **1** with the 2-chloroethanol.

via an intramolecular cyclization reaction, using the 2-chloroethyl 2-bromo-2-perfluoroalkylethanoate substrate with the zinc.

2. Results and discussion

The 2-bromo-2-perfluoroalkyl acids **1** [51] are transformed into the corresponding 2-chloroethyl 2-bromo-2-perfluoroalkylethanoates **2**, in good yields, by an alcoholysis reaction with the 2-chloroethanol in acidic medium for 48 h (Scheme 1).

Compounds **2** are reacted with activated zinc for 6 h at 70°C, in tetrahydrofuran (THF), then hydrolysed. The reaction leads exclusively to the corresponding 2-chloroethyl 2-perfluoroalkylethanoates **3**, resulting from the reduction reaction of the bromine atom in middle yields (Scheme 2).

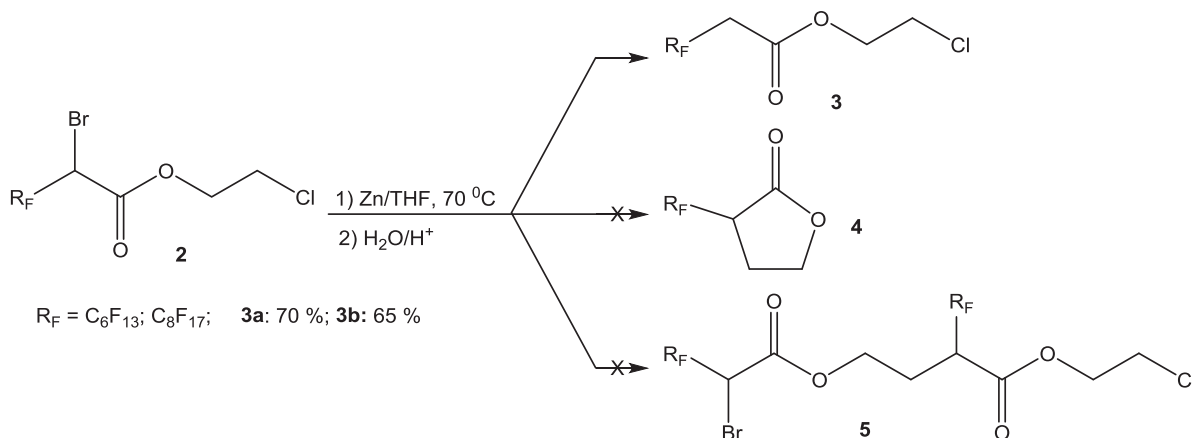
The zinc atom inserts between carbon and bromine atoms, yielding the bromozincic organometallic intermediate. This nucleophilic intermediate is unable to act on the electrophilic chloro-substituted carbon atom. So, the expected perfluoroalkylated lactones **4** are not obtained even by modifying some reaction conditions (zinc amount, temperature and reaction duration).

This result may be explained on the basis of the weak bromozincic carbon atom (CHZnBr) nucleophilicity [62] to act on the chloro-substituted carbon atom, neither via an intramolecular nucleophilic substitution (S_{Ni}) to produce the heterocyclic five-membered lactones **4**, nor via an intermolecular nucleophilic substitution reaction to yield the dimeric compounds **5** (Scheme 2).

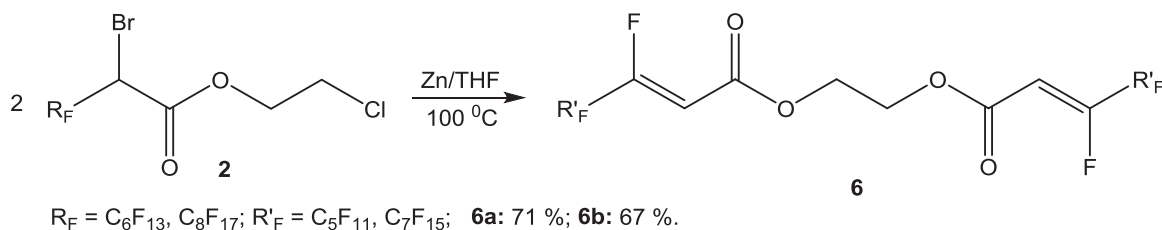
The reaction of the 2-chloroethyl 2-bromo-2-perfluoroalkylethanoate **2** in the presence of a zinc excess, in THF at 100 °C for 48 h, conduce to the symmetrical bis(vinyl perfluoroalkyl ester) **6** (Scheme 3) as unique product, resulting from a *trans* BrZnF elimination reaction, followed by an ethane molecule elimination than an O-alkylation (Scheme 5).

The formation of the esters **3** as well as the dimeric compounds **6** show that the reaction involves in all cases the bromozincic intermediate **A** (Scheme 4). While fluorine atom elimination reaction is not well frequent, because strength and the short length of the carbon-fluorine covalent single bond, such reaction occurs on compounds **2**. The high and prolonged heat, of perfluoroalkyleted bromoester **2**, conduces exclusively, through an elimination reaction of (BrZnF) to the *Z* geometrical configuration isomer **B**, [63] highly stabilized from a conjugated system. The geometrical structure of the bromozincic intermediate **A** (Scheme 4) is in favour of a *trans* elimination, because the β -carbon atom is bonded to two fluorine atoms ($\text{CF}_{2\alpha}$). So, regardless the absolute configuration of the α -carbon atom, in both cases, there is always a fluorine atom in the *trans* position to the ZnBr living group (Scheme 4).

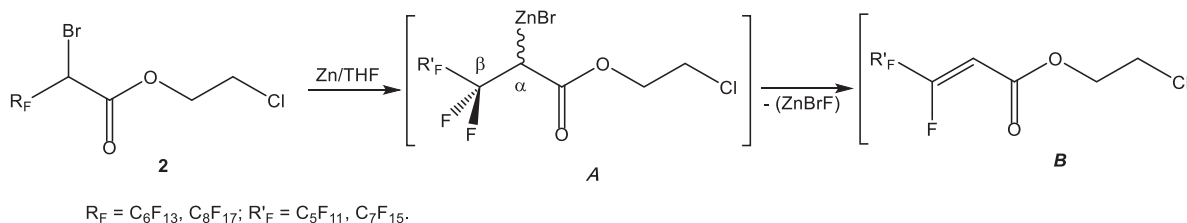
The presence of a zinc excess in the reaction with the ester **2**, in the above-described reaction conditions, gives rise also to a second zinc insertion reaction between chlorine and carbon atoms. As a result, the intermediate **B** transforms into the intermediate **C** (Scheme 5). This latter transforms in its turn into the corresponding ionic acid salt intermediate **C'**, via an elimination reaction of an ethene molecule.



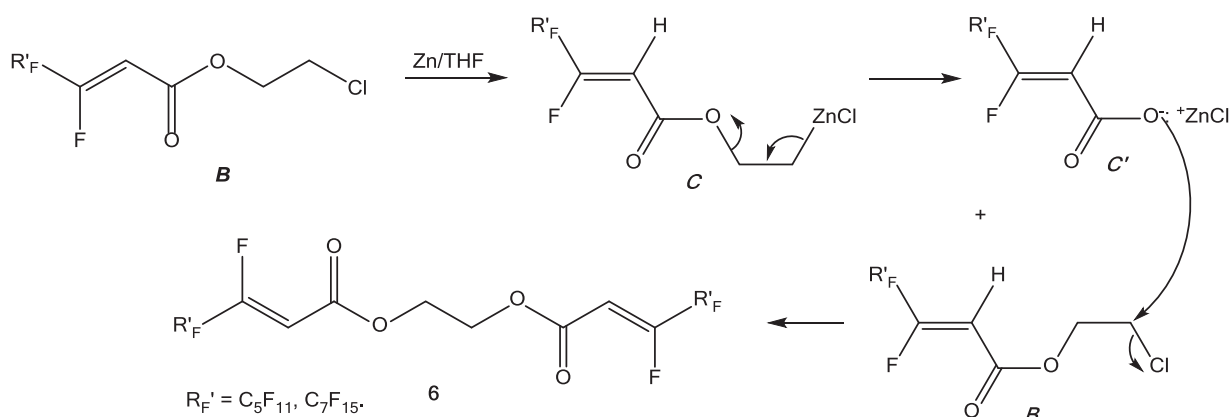
Scheme 2. Action of zinc on the 2-chloroethyl 2-bromo-2-perfluoroalkylethanoates **2** at 70°C.



Scheme 3. Effect of a zinc excess on the 2-chloroethyl 2-bromo-2-perfluoroalkylethanoates **2** at 100°C.



Scheme 4. *trans* elimination of BrZnF species from compounds **2** at 100°C.



Scheme 5. Formation of the bis(vinyl perfluoroalkyl ester) ethoxyde **6**.

The anionic species intermediate **C'** acts on the intermediate **B**, yielding the corresponding symmetric bis(vinyl perfluoroalkyl) ethoxide compound **6** (Scheme 5).

The NMR spectroscopic data confirm the obtained structures. In particular, the ^1H NMR spectra show a doublet at 6.05 ppm, corresponding to the two enonic protons ($=\text{CH}$), coupled with the single vinylic fluorine atom ($=\text{CF}$, $^3J_{\text{HFtrans}} \sim 31$ Hz). The ^{19}F NMR spectra are in agreement with the ^1H NMR spectra and show a signal corresponding to two vinylic fluorine atoms at -108 ppm. The ^{13}C NMR spectra show a singlet at ~ 106 ppm, attributed to the enonic ($\text{HC}=\text{C}$) carbon atoms and a doublet of multiplet at 152 and 156 ppm, relative to fluorinated vinylic carbon atoms ($=\text{CF}$) with a coupling constant $^1J_{\text{FC}} \sim 285$ Hz.

3. Conclusion

The 2-chloroethyl 2-bromo-2-perfluoroalkylethanoates **2** are synthesized by an alcoholysis reaction of 2-bromo-2-perfluoroalkyl acids **1** in the 2-chloroethanol. Reacted with the zinc in middle conditions, the

bromo-esters **2**, yield the bromine atom reduction products **3** and no trace of the desired perfluoroalkyl lactones **4** is detected. Such results justify again the low reactivity of the zincic carbon atom to act as a nucleophile on the primary chloro-substituted carbon atom. The prolonged heat of the same reagents **2** with a zinc excess, yield the bis(vinyl perfluoroalkyl) oxyethylene **6** as exclusive products.

The highly fluorinated obtained compounds 2-perfluoroalkyl acids **3** and bis(vinyl perfluoroalkyl ester) ethoxides **6**, among many others applications, may have interesting surface properties [64–66] and can also undergo some interesting reactions, particularly in the synthesis of organofluorinated heterocyclic compounds [67,68].

4. Experimental

^1H , ^{13}C and ^{19}F NMR spectra were realized in CDCl_3 on a Bruker AC 300 spectrometer at 300, 75 and 282 MHz, respectively, for ^1H , ^{13}C (TMS) and ^{19}F (C_6F_6). IR spectra were determined with PerkinElmer Paragon 1000 PC in CHCl_3 spectrometer. HRMS spectra were realized on a Finnigan MAT 95 mass spectrometer in

chemical ionization mode (CI). The used silicagel is of Merck 7734 type. The HRMS spectra were realized in the Institut National de Recherches et d'Analyses Physico-Chimiques (INRAP), Sidi Thabet 2020 Tunisia. The 2-bromo-2-perfluoroalkyl acids **1** were prepared according to the method described in Ref. [51].

4.1. Synthesis of 2-chloroethyl 2-bromo-2-perfluoroalkylethanoates **2**

A solution of 0.05 mmol of 2-bromo-2-perfluoroalkyl acid **1**, dissolved in 4 mL of 2-chloroethanol and 0.1 mL of concentrated H₂SO₄ is refluxed for 48 h. The mixture is cooled and diluted with 20 mL of water. The crude is extracted with (3 × 50 mL) diethyl ether, then dried over MgSO₄. The diethyl ether is evaporated and the residue is distilled, yielding ester **2** as yellowish liquid.

2-Chloroethyl 2-bromo-3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluorooctanoate: 2a

Bp = 64/0.01°C/mmHg; IR (CHCl₃): $\nu_{C=O}$ = 1756 cm⁻¹; ¹H NMR (CDCl₃), δ : 3.71 (t., 2H, CH₂-Cl), 4.50 (t., 2H, CH₂-O), 4.74 (q., 1H, CHBr); ¹³C NMR (CDCl₃), δ : 165.0 (d., C = O), 110.0–120.0 (m., C₆F₁₃), 65.0 (s., CH₂-O), 41.0 (s., -CH₂-Cl) and 40.0 (t., CHBr, ²J_{CF} = 63.1 Hz); ¹⁹F NMR (CDCl₃), δ : -84.3 (t., 3F, CF₃, ³J_{FF} = 7.9 Hz), -115.2 (m., AB_{Syst}, 2F, CF_{2 α} , ²J_{FF} = 279.2 Hz), -123.0 (m., AB_{Syst}, 2F, CF_{2 β} , ²J_{FF} = 296.3 Hz), -125.0 (m., 2F, CF_{2 γ}), -126.1 (m., 2F, CF_{2 δ}), -129.5 (m., 2F, CF_{2 ω}); HRMS: Calcd. for C₁₀H₅F₁₃BrClO₂: 517.89530, found 517.89559.

2-Chloroethyl 2-bromo-3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-heptadecafluorodecanoate: 2b

Bp = 89/0.01°C/mmHg; IR (CHCl₃): $\nu_{C=O}$ = 1753 cm⁻¹; ¹H NMR (CDCl₃), δ : 3.75 (t., 2H, CH₂-Cl), 4.51 (t., 2H, CH₂-O), 4.75 (q., 1H, CHBr); ¹³C NMR (CDCl₃), δ : 165.0 (d., C = O), 105.0–120.0 (m., C₆F₁₃), 65.4 (s., CH₂-O), 41.5 (s., -CH₂-Cl) & 40.2 (t., CHBr, ²J_{CF} = 62.6 Hz); ¹⁹F NMR (CDCl₃), δ : -84.4 (t., 3F, CF₃, ³J_{FF} = 9.1 Hz), -114.9 (m., AB_{Syst}, 2F, CF_{2 α} , ²J_{FF} = 272.4 Hz), -121.3 (m., AB_{Syst}, 2F, CF_{2 β} , ²J_{FF} = 298.4 Hz), -123.2 (m., 6F, 3CF_{2 γ , ϵ}), -125.2 (m., 2F, CF_{2 δ}), -128.4 (m., 2F, CF_{2 ω}); HRMS: Calcd. for C₁₂H₅F₁₇BrClO₂: 617.88900, Found 617.88943.

4.2. Synthesis of 2-chloroethyl 2-perfluoroalkylethanoate **3**

(4 mmol) of 2-chloroethyl 2-bromo-2-perfluoroalkylethanoate **2** dissolved in dry THF (20 mL) is added slowly via a dropping funnel to 0.26 g (4.0 mmol) of activated zinc. The mixture is heated at 70°C under vigorously stirring for 6 h. The reaction progress is controlled by TLC. At the end of the reaction, the solids are filtered off. The filtrate is hydrolysed with 15 mL of water then extracted with diethyl ether (3 × 50 mL). The solution is dried on Na₂SO₄, then the solvent is evaporated. The product is purified by chromatography using the mixture petroleum ether/

diethyl ether (70/30). The pure compounds **3** are obtained as yellowish liquids.

2-Chloroethyl 3,3,4,4,5,5,6,6,7,7,8,8,8-tridecafluorooctanoate: 3a

IR (CHCl₃): $\nu_{C=O}$ = 1750 cm⁻¹; ¹H NMR (CDCl₃), δ : 3.13 (t., 2H, CH₂, ³J_{HF} = 17.6 Hz), 3.72 (t., 2H, Cl-CH₂), 4.44 (m., 2H, O-CH₂); ¹³C NMR (CDCl₃), δ : 163.0 (s., C = O), 110.0–120.0 (m., C₆F₁₃), 65.8 (s., CH₂-O), 39.7 (s., -CH₂-Cl) and 35.0 (s., CH₂-CO); ¹⁹F NMR (CDCl₃), δ : -82.9 (t., 3F, CF₃), -113.1 (m., 2F, CF_{2 α}), -121.9 (m., 2F, CF_{2 β}), -123.6 (m., 4F, 2CF_{2 δ}), -127.0 (m., 2F, CF_{2 ω}); HRMS: Calcd. for C₁₀H₆F₁₃O₂Cl: 439.98487, found 439.98524.

2-Chloroethyl 3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-heptadecafluorodecanoate: 3b

IR (CHCl₃): $\nu_{C=O}$ = 1750 cm⁻¹; ¹H NMR (CDCl₃), δ : 3.18 (t., 2H, CH₂, ³J_{HF} = 17.6 Hz), 3.75 (t., 2H, CH₂-Cl), 4.48 (m., 2H, CH₂-O); ¹³C NMR (CDCl₃), δ : 163.3 (s., C = O), 110.0–120.0 (m., C₈F₁₇), 65.1 (s., CH₂-O), 40.7 (s., -CH₂-Cl) and 35.5 (t., 1C, CH₂-CF₂, ²J_{CF} = 86.1 Hz); ¹⁹F NMR (CDCl₃), δ : -82.7 (t., 3F, CF₃), -113.0 (m., 2F, CF_{2 α}), -121.1 (m., 6F, 3CF₂), -123.6 (m., 4F, 2CF₂), -126.9 (m., 2F, CF₂); HRMS: Calcd. for C₁₂H₆F₁₇O₂Cl: 539.97848, found 539.97877.

4.3. Synthesis of bis(allylperfluoroalkyl) ethoxide **6**

0.26 g (4 mmol) of activated zinc powder and 10 mL of dry THF under nitrogen atmosphere, (4 mmol) of 2-chloroethyl 2-bromo-2-perfluoroalkylethanoate **2** are placed in a 100 mL erlenmeyer. The mixture is vigorously stirred during 48 h at 100°C, then cooled, filtered and diluted with 100 mL of diethyl ether. The solution is washed with water (2 × 50 mL) then dried on MgSO₄. The solvents are evaporated and the obtained crude is purified on chromatographic column using the mixture petroleum ether/diethyl ether (70/30). Compounds **6** are isolated as yellowish solids.

(2Z,2'Z)-ethane-1,2-diyl bis(3,4,4,5,5,6,6,7,7,8,8,8-dodecafluorooct-2-enoate): 6a

Mp = 67°C; ¹H NMR (CD₃Cl₃), δ : 6.01 (d., 2H, 2CH = CF, ³J_{HF} = 31.1 Hz), 4.50 (s., 4H, 2CH₂-O); ¹³C NMR (CD₃Cl₃), δ : 160.9 (s., 2C, C = O), 152.3 and 156.2 (d.m., 2C, 2FC=, ¹J_{CF} = 285.1 Hz), 105.7 (s., 2C, 2CH=), 110.0–120.0 (m., 10C, C₇F₁₅), 65.0 (s., 2C, 2CH₂-O); ¹⁹F NMR (CDCl₃), δ : -82.6 (t., 6F, 2CF₃), -108.3 (m., 2F, 2CF=), -120.4 (m., 4F, 2CF₂), -124.4 (m., 8F, 4CF₂), -127.9 (m., 4F, 2CF₂); HRMS: Calcd. for C₁₈H₆F₂₄O₄: 741.98828, found 741.98853.

(2Z,2'Z)-ethane-1,2-diyl bis(3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-hexadecafluorodec-2-enoate): 6b

Mp = 85°C; ¹H NMR (CD₃Cl₃), δ : 6.08 (d., 2H, CF = CH, ³J_{HF} = 31.4 Hz), 4.60 (s., 4H, 2CH₂-O); ¹³C NMR (CDCl₃), δ : 163.5 (s., 2C, C = O), 152.3 and 156.2 (d.m., 2C, 2FC=, ¹J_{CF} = 286.7 Hz), 105.7 (s., 2C, 2CH=), 110.0–120.0 (m., 14C, C₇F₁₅), 65.1 (s., 2C, CH₂-O); ¹⁹F NMR (CDCl₃), δ : -82.5 (t., 6F, 2CF₃), -108.3 (m., 2F, 2CF=), -120.4 (m., 4F, 2CF₂), -123.6 (m., 8F, 4CF₂), -124.4

(m., 8F, 4CF₂), −127.8 (m., 4F, 2CF₂); HRMS: Calcd. for C₂₂H₆F₃₂O₄: 941.97551, Found 941.97589.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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